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Master's degree program in Data Science

Reliability of Similarity Metrics in Detecting Plagiarized Texts: A Comparison with Human Judgment

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Academic Year 2023-2024

Abstract

This study investigates the reliability of automated similarity metrics in assessing textual similarity compared to human evaluations. Using a dataset of original and paraphrased texts, we analyze how well different computational similarity measures (that is cosine similarity, BLEU, ROUGE-L and METEOR) correspond to human perception of similarity. Furthermore, we explore the impact of paraphrasing, translation, and minor textual modifications on the effectiveness of these metrics in detecting plagiarism. Through a combination of quantitative analysis and human evaluation, we aim to determine whether existing similarity measures are sufficient for plagiarism detection or if they fail to capture nuanced cases of text transformation. Our findings will provide insights into the strengths and limitations of these metrics and contribute to the development of more robust tools for text similarity assessment.

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Chapter 1

Introduction

1.1 Background

Plagiarism has long been a concern in academia, journalism, and creative industries, but the digital era has transformed both the nature of plagiarism and the methods used to detect it. With vast amounts of information readily accessible online and AI-powered tools capable of generating sophisticated paraphrases, distinguishing between original work and borrowed content has become increasingly complex. Traditional notions of plagiarism, direct copying without attribution, no longer capture the full extent of modern text reuse. Today, subtle forms of plagiarism, such as structural paraphrasing and synonym substitution, challenge conventional detection methods and raise important questions about authorship and intellectual integrity.

Plagiarism, broadly defined, represents a violation of academic and scientific integrity [1]. It occurs when an individual presents someone else's ideas, words, or research findings as their own without proper acknowledgement. This encompasses not only direct copying but also more subtle forms of misappropriation, such as rewording or restructuring original content. The ethical and legal implications of plagiarism make its detection and prevention critical in academic, journalistic, and professional settings.

The formal recognition of plagiarism as a form of research misconduct emerged in the 1980s when institutions began distinguishing between responsible and irresponsible academic behaviour. Initially, plagiarism was defined narrowly as the theft of ideas

and uncredited copying. Over time, its definition expanded to include broader forms of intellectual misappropriation, leading to challenges in defining acceptable academic practices [1].

Today, plagiarism is classified as a type of *misappropriation*, which includes presenting another's documented words or ideas as one's own without proper attribution. This extends beyond published works to encompass unpublished research, confidential documents, and even peer-reviewed materials. The U.S. federal government defines plagiarism as "the appropriation of another person's ideas, processes, results, or words without giving appropriate credit," emphasizing that it applies throughout the research lifecycle—from proposal to publication [1].

Efforts to combat plagiarism have historically relied on exact text matching, flagging instances where a sequence of words appears in multiple sources. While effective for identifying verbatim copying, these approaches fall short when dealing with more nuanced forms of textual modification. As a result, modern detection systems increasingly rely on similarity metrics that assess not just lexical overlap, but also semantic equivalence. Metrics such as BLEU, ROUGE, and METEOR provide computational methods for evaluating text similarity, each with distinct strengths and weaknesses in capturing meaning beyond surface-level text matching.

Understanding the limitations of these automated metrics is critical, as they influence how plagiarism detection systems classify reworded content. This study investigates the reliability of widely used similarity measures, comparing their outputs against human judgments to determine how effectively they distinguish between original and paraphrased texts. By exploring how paraphrasing, translation, and minor textual modifications impact similarity scores, we want to see whether existing computational approaches align with human perception and, ultimately, whether they can serve as robust tools in the fight against plagiarism.

1.2 Plagiarism Detection

Plagiarism detection methodologies can be categorized into three primary types:

1. **Verbatim Plagiarism**

2. Paraphrased Plagiarism

3. Cross-Language Plagiarism

Each type requires distinct computational approaches for effective identification.

1.2.1 Verbatim Plagiarism Detection

Verbatim plagiarism, the most straightforward form, involves copying text word-for-word without any modification or citation. The detection of such plagiarism can be efficiently performed using these techniques:

- **N-gram Analysis:** This method segments text into overlapping sequences of words or characters (n-grams) and compares them across documents. It is effective for detecting direct matches but struggles with paraphrased or slightly altered text[10];
- **Hashing Techniques:** Hash-based methods generate a unique hash value for text segments, enabling rapid comparison between documents. While efficient, this approach fails to detect structural or semantic modifications;
- **Vector Space Models (VSM):** This method represents text documents as vectors in a multi-dimensional space, enabling similarity measurements based on cosine similarity or other distance metrics [10].

1.2.2 Paraphrased Plagiarism Detection

Paraphrasing plagiarism is more sophisticated, involving the rewording or restructuring of an original text while maintaining its core meaning. This form of plagiarism is particularly challenging to detect, as it often evades traditional keyword-based approaches. Methods to address paraphrased plagiarism include:

- **Latent Semantic Analysis (LSA) and Explicit Semantic Analysis (ESA):** These techniques capture deeper semantic relationships between words and phrases, allowing for similarity detection beyond exact matches[10];
- **Word Embeddings:** Approaches like Word2Vec and FastText map words to high-dimensional vector spaces, where semantically similar words are positioned closer together. This enables better identification of paraphrased content;

- **Paraphrase Identification:** Machine learning models trained on paraphrase datasets can detect variations in sentence structure while preserving meaning, making them effective tools for identifying disguised plagiarism [24].

1.2.3 Cross-Language Plagiarism Detection

Cross-language plagiarism, also known as *translingual plagiarism*, occurs when a text is translated from one language to another without proper attribution. Since traditional methods struggle with detecting translated text, specialized techniques are employed:

- **Reverse Translation:** This technique involves translating the suspicious text back into the original language and comparing it with potential source documents. While effective, it is dependent on translation accuracy;
- **Interlingual Analysis:** This method analyzes linguistic interference left behind during translation, such as unnatural phrase structures or borrowed syntax, to identify possible plagiarism [24];
- **Multilingual Word Embeddings:** Extensions of word embedding techniques allow cross-language comparisons by mapping words into a shared semantic space, making it possible to detect similarities even when the text has been translated [10].

1.2.4 Advancements in Plagiarism Detection

Traditional plagiarism detection methods, while effective for exact matches, struggle with more complex cases of paraphrasing, stylistic modifications, and cross-lingual transformations. To address these challenges, modern detection techniques use advanced machine learning and deep learning models:

- **Transformer-based Models (BERT, RoBERTa, T5):** These models generate context-aware vector representations of text, improving semantic similarity detection. BERT, for instance, excels at capturing nuanced meaning, making it useful for identifying paraphrased and subtly altered content.
- **Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM):** These models analyze text sequences over time, helping detect stylistic inconsisten-

cies and paraphrased content.

- **Stylometric Analysis:** By analyzing an author’s writing style (word frequency, syntax, sentence structure), stylometric methods can flag inconsistencies that suggest potential plagiarism [10].

1.2.5 Historical Evolution of Plagiarism Detection

Plagiarism detection has undergone a significant transformation over the years. Initially, plagiarism was identified manually by experts through careful examination of texts. With the advent of digital technology, automated tools such as Turnitin and CrossCheck emerged, providing more systematic and large-scale solutions. More recently, artificial intelligence (AI) and machine learning techniques have further advanced the field, enabling deeper semantic analysis and detection of paraphrased content. These advancements have allowed for more efficient and accurate detection, but they also introduce new challenges, as sophisticated methods for content manipulation evolve.

1.2.6 Ethical and Legal Implications of Plagiarism

Plagiarism is not only a technical problem but also an ethical and legal concern. In academia, plagiarism undermines the integrity of scholarly work and can lead to severe consequences, including retraction of papers, academic penalties, and reputational damage. In journalism and content creation, plagiarism can result in legal actions and loss of credibility. Notable cases of plagiarism scandals in research and media have demonstrated the importance of robust detection mechanisms to uphold academic and professional integrity.

1.2.7 Challenges in Modern Plagiarism Detection

Despite advancements in plagiarism detection, modern tools still face several challenges. One of the primary difficulties is detecting AI-generated paraphrases, where text is modified at a syntactic level while retaining its original meaning. Adversarial attacks against plagiarism detectors further complicate the task, as individuals deliberately manipulate text to evade detection. Moreover, translation-based obfuscation techniques, where a text is translated into multiple languages and back to the original, present additional

difficulties for traditional detection systems.

1.2.8 Role of Large Language Models (LLMs) in Plagiarism and Detection

Large language models (LLMs) such as GPT-4 have revolutionized text generation, making it easier than ever to produce paraphrased and original-looking content. While these models enable creativity and efficiency, they also raise concerns about their role in facilitating plagiarism. On the other hand, LLMs can also be used for advanced plagiarism detection, as they are capable of capturing deeper semantic similarities beyond simple lexical matches. The question of whether AI-generated paraphrases should be considered plagiarism remains an ongoing debate, especially as AI-assisted writing tools become mainstream.

1.2.9 Human vs. Machine Perception of Similarity

An important aspect of plagiarism detection is the difference between human and machine perception of similarity. Humans assess similarity based on context, meaning, and intent, whereas computational models rely on lexical and semantic features. This gap often leads to discrepancies in detection results, where some automated systems may flag benign cases while missing more subtle forms of plagiarism. Evaluating the reliability of computational similarity measures in comparison to human judgment is a crucial research question in this domain.

1.2.10 Cross-Domain Applications of Text Similarity

Beyond plagiarism detection, text similarity analysis has applications in various fields, including information retrieval, authorship verification, and fake news detection. In authorship attribution, for instance, text similarity measures help determine whether a piece of writing belongs to a specific author. Similarly, in misinformation detection, semantic similarity analysis can reveal manipulated content or reworded false information. These interdisciplinary applications further highlight the importance of developing robust and reliable similarity assessment techniques.

These aspects collectively frame the significance of studying text similarity and pla-

gicism detection. In the subsequent sections, we explore these challenges in more depth and propose methodologies to improve automated similarity assessments and plagiarism detection frameworks.

1.3 Text Similarity Metrics

In the context of plagiarism detection and text similarity analysis, various automated metrics have been developed to quantify the similarity between texts. These metrics serve as computational tools to assess the extent to which a given text is a recognizable variant of an original. However, the reliability of these metrics compared to human judgment remains an open question.

This section explores several widely used similarity measures, including cosine similarity, BLEU, ROUGE, and BERT-based embeddings, analyzing their theoretical foundations, computational mechanisms, and effectiveness in detecting text transformations such as paraphrasing, translation, and minor textual modifications.

1.3.1 Cosine Similarity

Cosine similarity is a fundamental metric for assessing the similarity between two text vectors. It measures the similarity between two non-zero vectors by calculating the cosine of the angle between them[6]. Similarity measure calculates the distance between data objects based on their feature dimensions in a dataset and a smaller distance indicates a higher similarity, while a larger distance indicates a lower similarity[6]. It is commonly used in information retrieval and text clustering[23].

Mathematical Definition: The cosine similarity between two vectors A and B is computed as:

$$\text{cosine similarity} = \frac{A \cdot B}{\|A\| \|B\|} \quad (1.1)$$

where:

- $A \cdot B$ represents the dot product of vectors A and B .
- $\|A\|$ and $\|B\|$ are the Euclidean norms of the respective vectors.

Range of Values:

- 1 indicates identical vectors.
- 0 signifies orthogonality, implying no correlation.
- -1 denotes opposite vectors.

To calculate cosine similarity, documents or texts are converted into vectors. These vectors can represent the frequencies of words in documents (TF-IDF) or other attributes. Cosine similarity can be seen as a method of normalizing the length of the document during comparison and one of the main advantages of cosine similarity is its low computational complexity, especially for sparse vectors, since only the non-zero coordinates need to be considered[31].

Cosine similarity is particularly useful in vector space models such as TF-IDF and word embeddings. It efficiently captures the semantic similarity between documents .

1.3.2 BLEU (Bilingual Evaluation Understudy)

BLEU is an automated evaluation metric originally designed for machine translation but also applicable to text similarity analysis[20]. It assesses translation quality by comparing n-grams in candidate texts to reference translations. The core idea of the BLEU metric is that a machine translation is better the closer it is to a professional human translation[20].

Key Features[30]:

- BLEU evaluates quality by comparing the output of a machine translation system with one or more human reference translations;
- Uses a brevity penalty to discourage excessively short outputs;
- BLEU is based on comparing n-grams (sequences of words) between the candidate text and the reference translations. It calculates the modified precision of n-grams, taking into account the maximum frequency with which each word appears in the reference translations. For the modified precision of unigrams, the occurrences of each candidate word found in any reference translation are counted, limiting the maximum count of each word based on the maximum count in any of the reference translations;
- BLEU score is a number between 0 and 1, where values closer to 1 indicate greater

similarity to the reference translations and therefore higher quality of the machine translation. A score of 1 is rare even for human translations, as it would imply that the candidate translation is identical to a reference one.

Formula[7]:

$$\text{BLEU} = \text{BP} \times \exp \left(\sum_{n=1}^N w_n \log(p_n) \right) \quad (1.2)$$

where:

- BP is the brevity penalty.
- p_n represents n-gram precision.
- w_n are the n-gram weights.

While BLEU is effective in identifying verbatim text matches, it struggles with many limitations like it is not suitable for languages without word delimiters or it may not always reflect an improvement in translation quality if the score increases and it is dependent on the tokenization technique. Therefore, BLEU has some problems with synonym substitution and sentence reordering, limiting its reliability in plagiarism detection[20][30].

1.3.3 ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

ROUGE is a suite of metrics designed to evaluate text summarization and machine translation quality. Unlike BLEU, which emphasizes precision, ROUGE prioritizes recall, making it better suited for detecting information retention[33][18].

Variants of ROUGE:

- **ROUGE-N:** Measures n-gram overlap;
- **ROUGE-L:** Uses longest common subsequence (LCS) to assess similarity;
- **ROUGE-S:** Incorporates skip-bigram co-occurrence.

In particular, ROUGE-L is a text similarity metric that evaluates the longest common subsequence (LCS) between a candidate text and a reference text. Unlike ROUGE-N (which relies on exact n-gram matches), ROUGE-L captures sentence-level structure

similarity by considering the longest overlapping sequence of words that appear in order, though not necessarily consecutively[18].

ROUGE scores range between 0 and 1. A higher score indicates greater similarity between the automatically generated text and the reference text.

ROUGE-L is computed using three components: recall, precision and F1-score, based on the LCS[18]:

- **Longest Common Subsequence (LCS):** Given a reference sequence RR and a candidate sequence CC , the LCS is the longest sequence of words that appear in both texts in the same order, though not necessarily consecutively;

- **Recall (R-LCS):**

$$R_{LCS} = \frac{LCS(C, R)}{|R|} \quad (1.3)$$

Measures how much of the reference text is covered by the candidate text;

- **Precision (P-LCS):**

$$P_{LCS} = \frac{LCS(C, R)}{|C|} \quad (1.4)$$

Measures how much of the candidate text matches the reference text;

- **F1-Score (Rouge-L F1):**

$$F1_{LCS} = \frac{(1 + \beta^2) \cdot R_{LCS} \cdot P_{LCS}}{R_{LCS} + \beta^2 P_{LCS}} \quad (1.5)$$

The F1-score balances recall and precision. The default value of β is usually set to a very high number like 0.8, making recall and precision equally important.

ROUGE is often compared with other metrics, such as BLEU and METEOR, which are used for similar purposes in natural language processing. While BLEU focuses on the precision of the translation, ROUGE focuses on the recall of information from the reference text.

ROUGE is widely used in various fields, like automatic text summarisation, where it evaluates the quality of automatically produced summaries by comparing them with reference summaries, and machine translation, where it evaluates the accuracy and completeness of automatically produced translations.

1.3.4 METEOR (Metric for Evaluation of Translation with Explicit ORdering)

METEOR (Metric for Evaluation of Translation with Explicit ORdering) is an automatic evaluation metric originally designed for machine translation but also applicable to measuring textual similarity[3]. Unlike traditional n-gram overlap metrics such as BLEU and ROUGE, METEOR incorporates multiple word-matching strategies, making it more sensitive to meaning preservation in paraphrased and translated texts. Given that this study investigates the reliability of automated similarity metrics against human evaluation, METEOR’s approach aligns well with the focus on paraphrasing, translation, and minor textual modifications.

METEOR is based on the harmonic mean of precision and recall[32]:

- **Precision (P)**: The fraction of words in the candidate text that match the reference text.
- **Recall (R)**: The fraction of words in the reference text that appear in the candidate text.

Unlike BLEU, which prioritizes precision, METEOR emphasizes both recall and precision and it combines them into a single score, aligning with human perception that favours semantic completeness[13].

The standard F1-score is defined as the harmonic mean of precision and recall:

$$F_{\text{mean}} = \frac{(1 + \beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R} \quad (1.6)$$

where β is a parameter that controls the relative weight of recall compared to precision.

However, METEOR does not use this standard F1-score formula. Instead, it applies a weighted harmonic mean with an adjustable parameter α that explicitly controls the balance between precision and recall[11]:

$$F_{\text{mean}} = \frac{P \cdot R}{\alpha P + (1 - \alpha)R} \quad (1.7)$$

where $P = \frac{m}{c}$ and $R = \frac{m}{r}$, with:

- m being the number of mapped unigrams,
- c the total number of unigrams in the candidate translation, and
- r the total number of unigrams in the reference translation.

Unlike the traditional F1-score, METEOR’s formulation places more emphasis on recall by setting α to a value lower than 0.5 (typically around 0.9), making recall more influential in determining the final score.

Additionally, METEOR includes a penalty function to account for word order and fluency[11]:

$$\text{Penalty} = \gamma \left(\frac{c_m}{m} \right)^\beta \quad (1.8)$$

where:

- c_m is the number of contiguous matching chunks,
- γ is a parameter controlling the fragmentation penalty, and
- β adjusts the penalty’s shape.

Finally, the METEOR score is computed as:

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty}) \quad (1.9)$$

This formulation ensures that METEOR not only considers the precision-recall balance but also accounts for translation fluency by penalizing fragmented matches.

METEOR extends beyond exact word matching by incorporating[32]:

- **Stemming:** Matching words based on their root forms (e.g., “run” vs. “running”).
- **Synonym Matching:** Using semantic similarity (e.g., “big” vs. “large”).
- **Paraphrase Matching:** Recognizing reworded phrases, which makes METEOR more resilient to paraphrasing than BLEU or ROUGE.

METEOR is really useful for this study because this metric uses stemming and synonym matching which provide an advantage over rigid n-gram metrics like BLEU and ROUGE. In particular, unlike BLEU, which only considers exact word matches, METEOR accounts for meaning variations, making it more aligned with human judgments.

1.3.5 Comparative Analysis of Metrics

While each metric provides valuable insights into text similarity, they also have inherent limitations:

- **Cosine similarity** is widely used for measuring text similarity in vectorized representations, particularly in large-scale applications. However, research has shown that cosine similarity may yield arbitrary and unreliable results when applied to learned embeddings[26]. While it is effective for comparing term frequency and distribution, it does not always reflect meaningful semantic relationships. The Netflix study highlights that the issue arises not from cosine similarity itself but from the degree of freedom in learned embeddings, which can lead to misleading similarity scores. In contrast, alternative approaches such as the unnormalized dot product often produce more stable and well-defined results[9].
- **BLEU (Bilingual Evaluation Understudy)** is widely used in text similarity and machine translation evaluation. It measures similarity based on n-gram precision, making it effective for assessing direct word overlap. However, BLEU struggles with paraphrased content and synonym variations, as it does not consider meaning beyond exact matches[2][20].
- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation)** is designed to evaluate text summarization by focusing on recall—how much of the reference text is retained in the candidate text. Unlike BLEU, ROUGE places greater emphasis on recall, making it more suitable for detecting coverage rather than precision[18]. However, like BLEU, it does not incorporate deep semantic understanding, and its reliance on n-gram overlap makes it vulnerable to missing paraphrased content that conveys the same meaning in a different structure[12].
- **METEOR (Metric for Evaluation of Translation with Explicit ORdering)**

improves upon BLEU by incorporating stemming, synonym matching, and a fragmentation penalty. Unlike BLEU, which strictly matches exact words, METEOR allows for flexible matching through stemmed words and synonyms, increasing its ability to recognize paraphrased content. The fragmentation penalty accounts for the fluency of a candidate text by penalizing disjoint matches, thus promoting better linguistic coherence. However, METEOR has some limitations as well, for instance, the weights assigned to different measures in the calculation of the METEOR score may not always be optimal or representative of the desired criteria for evaluating translation quality and the algorithm behind the METEOR score is relatively complex and computationally intensive, which can make it slower and more resource-intensive than other metrics[3][14].

A comprehensive plagiarism detection system should integrate multiple similarity measures to take care of different types of text transformations. By incorporating human evaluations into the assessment framework, we can better understand the reliability of these metrics and refine them for real-world applications in detecting text modifications.

1.4 Thesis Objective

The primary objective of this thesis is to assess the reliability of automated similarity metrics in evaluating textual similarity compared to human judgments. Computational similarity measures—such as cosine similarity, BLEU, ROUGE, and BERT-based embeddings—are widely used in tasks such as paraphrase detection, plagiarism detection, and authorship verification. However, their effectiveness in capturing human-perceived similarity remains an open question, particularly when texts undergo systematic transformations.

To investigate this, we construct a framework that generates text variations using controlled modifications, including:

- **Grammar modifications:** Altering sentence structure while preserving meaning.
- **Contextual paraphrasing:** Utilizing transformer-based models to generate semantically equivalent yet structurally different sentences.
- **Multilingual translation and retranslation:** Evaluating how translations affect

similarity scores by translating text into another language and back.

We use the Microsoft Research Paraphrase Corpus (MRPC) as a benchmark dataset, applying these transformations to its sentence pairs. The goal is to analyze how well similarity metrics capture meaning preservation across different types of text perturbations.

To validate the reliability of these metrics, we incorporate human evaluation. A set of perturbed texts is presented to human annotators, who assess their similarity to the original texts. By comparing human judgments with automated similarity scores, we quantify the alignment between computational and human perception. We further analyze precision and recall to measure the effectiveness of these metrics in distinguishing paraphrases from non-paraphrases.

This thesis aims to answer the following research questions:

1. How well do automated similarity metrics (e.g., BLEU, ROUGE-L, METEOR) correlate with human judgments of textual similarity?
2. Can similarity metrics effectively distinguish between paraphrased and non-paraphrased texts using statistical correlation analysis?
3. How well do similarity metrics perform in binary classification tasks for detecting textual similarity, as evaluated through Receiver Operating Characteristic (ROC) analysis?
4. What are the optimal threshold values for different similarity metrics when classifying texts as similar or dissimilar?

By addressing these questions, this research contributes to a better understanding of the strengths and limitations of similarity metrics in text analysis. The findings have implications for applications in authorship verification, plagiarism detection, and natural language processing evaluation.

Chapter 2

Methodology

2.1 Overview

This chapter outlines the methodological framework employed to evaluate the reliability of automated similarity metrics in assessing text similarity compared to human perception. The approach involves computing various similarity metrics on a dataset of paraphrased and translated sentences and comparing these metrics with human evaluations of similarity. The ultimate goal is to assess how well these metrics align with human perceptions of text similarity and to identify which metrics are most reliable for detecting paraphrased variants of original texts.

2.2 Dataset and Translation Process

The study utilizes a dataset of 100 sentences from the Microsoft Research Paraphrase Corpus (MRPC). This dataset us from a text file containing 5800 pairs of sentences which have been extracted from news sources on the web, along with human annotations indicating whether each pair captures a paraphrase/semantic equivalence relationship. No more than 1 sentence has been extracted from any given news article.[19] These sentences were paraphrased using a layered transformation approach and then translated into 10 different languages using two popular translation services, Google Translate and DeepL. The languages included high-resource languages such as French, Spanish, and German, as well as medium-resource languages like Turkish and Japanese. This process resulted

in a total of 2000 translated sentences. The translated sentences were then retranslated back into English to analyze how much meaning was preserved.

The paraphrased and translated texts were evaluated using various similarity metrics to measure how closely they resemble the original sentences. The goal of this dataset creation process was to generate textual variations that could be used to test the effectiveness of similarity metrics and to assess how these modifications affect human perception of similarity.

2.3 Layered Paraphrasing Approach

The paraphrasing technique developed in this thesis uses a layered approach to introduce controlled modifications in the text while preserving meaning. The key components of this approach are as follows:

1. **Grammar Transformations:** This layer applies minor grammatical changes to the text without altering its overall meaning. Transformations include converting verb tenses (e.g., present to past) and replacing auxiliary verbs (e.g., *is* to *was*).
2. **Contextual Paraphrasing:** In this layer, sentences are paraphrased using a T5-based paraphrasing model. This step rephrases the text while retaining its core meaning, embedding additional variability in wording and sentence structure.
3. **BERT-Based Contextual Replacements:** Specific words or phrases within the text are replaced with semantically similar alternatives using a BERT masked language model. For example, the word *rapid* might be replaced with *swift* based on the context of the sentence. This layer ensures that the changes are grammatically valid and semantically coherent.

By combining these layers, the final paraphrased text incorporates multi-level transformations that make it distinct yet semantically similar to the original text.

2.4 Similarity Metrics

To evaluate the similarity between the original and translated sentences, the following metrics, seen in 1.3, were computed:

- **Cosine Similarity**: Measures the cosine of the angle between two vectors of term frequencies, providing a measure of similarity based on word overlap;
- **BLEU (Bilingual Evaluation Understudy)**: Assesses the similarity by comparing n-grams between the original and translated texts, focusing on precision;
- **ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation - Longest Common Subsequence)**: Evaluates similarity based on the longest common subsequence, emphasizing recall;
- **METEOR (Metric for Evaluation of Translation with Explicit ORdering)**: Combines precision, recall, and alignment between the original and translated texts, providing a balanced measure of similarity.

2.5 Selection of Sentences for Human Evaluation

To ensure a thorough and insightful evaluation, a subset of 94 sentence pairs was carefully selected from the 2000 paraphrased and translated sentences. This selection was based on a systematic approach that considered similarity metric values across various metrics, including Cosine Similarity, BLEU, ROUGE-L, and METEOR. The goal was to capture a diverse range of similarity scores, reflecting variations in translation quality, paraphrase strength, and overall similarity.

The selection process involved defining specific criteria to identify sentence pairs with interesting characteristics in their similarity metrics. These criteria included combinations such as low Cosine Similarity with high METEOR scores and significant differences between ROUGE-1 and ROUGE-L scores. A Python script was used to automate the selection process based on these criteria.

For a detailed explanation of the selection criteria and the methodology used, please refer to section 3.5.1. This approach ensured that the dataset was representative and diverse, providing a good foundation for the human evaluation process.

2.6 Human Evaluation

To understand how well the similarity metrics align with human perceptions of text similarity, a survey was conducted using Google Forms. Participants were asked to evaluate the similarity of the 94 selected sentence pairs. In particular the choices that could be ticked were:

- **Completely different** (The two sentences have no meaning in common.);
- **Slightly similar** (There are some shared words, but the meaning is mostly different.);
- **Moderately similar** (The meaning is somewhat preserved, but there are significant changes.);
- **Very similar** (The meaning is mostly the same with minor wording differences.);
- **Almost identical** (The two sentences convey exactly the same meaning.).

The human ratings were then averaged to generate a reference similarity score for each pair, which was compared with the computed similarity metrics.

This evaluation was designed to assess whether computational similarity measures (such as cosine similarity, BLEU, and BERT-based embeddings) correspond to human judgments when determining whether a given text is a recognizable variant of an original text.

2.7 Analysis of Results

The analysis involved computing the correlation between human ratings and the similarity metrics. This allowed for an assessment of which metrics best capture human perceptions of text similarity. The key analytical steps included:

- Computing Pearson and Spearman correlation coefficients between human similarity ratings and metric scores.
- Analyzing the mean similarity scores for each metric across different languages and translation services to identify trends and patterns.

- Investigating cases where automated metrics and human evaluations diverged significantly, to understand potential weaknesses in current similarity metrics.

The results provided insights into the reliability of similarity metrics and their effectiveness in detecting paraphrased and translated variants of texts. Understanding these relationships is crucial for evaluating the robustness of similarity detection methods in multilingual and paraphrased contexts.

2.8 Languages and Translation/Retranslation Process

To evaluate the robustness of the paraphrase, the text undergoes translation and retranslation across multiple languages using different translation services. This process simulates real-world scenarios where original texts might be modified by users or systems operating in multilingual contexts. The details of this process are as follows:

- **Languages:** The study includes a diverse set of languages to test the watermark’s resilience across linguistic variations. High-resource languages such as French, Spanish, and German are included for their well-established translation systems. Additionally, medium-resource languages like Turkish and Japanese are incorporated to evaluate the watermark’s performance under less optimal translation conditions. This selection ensures a comprehensive analysis of linguistic factors affecting watermark detectability.
- **Translation Services:** Two popular translation services, Google Translate and DeepL, are used to perform the translation and retranslation processes. These services employ distinct neural machine translation architectures, providing a robust test of the watermark’s resilience. Google Translate is widely known for its extensive language support and general-purpose translations, while DeepL is recognized for its high-quality translations, particularly in European languages.

2.8.1 Languages

In order to understand potential differences between the performance of romance languages and languages from other families in this thesis, we need to understand in what they differ.

Romance languages, such as Spanish, French, and Italian, derive from Vulgar Latin, the spoken language of the Roman Empire. Languages like Turkish and Japanese, however, belong to different language families and have distinct evolutionary histories. This difference in origin is reflected in their grammatical and lexical structures[27].

Regarding origin and language families, Romance languages belong to the Indo-European family, descending from Vulgar Latin. Turkish belongs to the Turkic language family, which is unrelated to Indo-European languages. Japanese is considered an isolate language, meaning no relationship with other language families has been proven[27][17].

In terms of morphological structure, Romance languages are fusional, where a single ending can express multiple grammatical functions (like gender, number, and case), as in the Latin word "*bonus*". Turkish is an agglutinative language, where suffixes are added sequentially to a word root, and each suffix has a specific grammatical function, as in the Turkish word "*evlerinizden*". Japanese has complex morphology, combining agglutination and other morphological processes, such as the use of classifiers[21].

Word order also differs significantly. Romance languages generally follow an SVO (Subject-Verb-Object) order in declarative sentences, though variations exist. For example: "The snake (S) bit (V) the man (O)." Both Turkish and Japanese follow an SOV (Subject-Object-Verb) order, where the verb appears at the end of the sentence[5].

Regarding vocabulary, Romance languages share a common lexical base derived from Latin, though they have developed regional specificities and borrowed words from other languages over time. Turkish and Japanese have very different lexicons from Romance languages, influenced by their own histories and cultures[5].

To wrap it up, while Romance languages share a Latin origin and certain structural and lexical characteristics, Turkish and Japanese belong to distinct language families, with grammatical structures, word order, and vocabulary that significantly differ from Romance languages.

2.8.2 Google Translate

Google Translate is a multilingual machine translation service by Google that translates text, documents, and websites between languages[29]. As of January 2025, it supports 249 languages and language varieties, serving over 500 million users daily and translating more than 100 billion words per day[16]. For more information, visit the official Google Translate website: [Google Translate](https://www.google.com/translate).

The service offers text, website, document, voice, and conversation translation through web interface, mobile apps (Android/iOS), and developer API.

Translation methodology evolved from Statistical Machine Translation (SMT) to Neural Machine Translation (NMT) in 2016. The current system uses Transformer-based deep learning networks to translate entire sentences at once, aiming to encode sentence semantics rather than phrase-by-phrase translation[15].

Key features include things like pronunciation guides and dictionary functions or automatic language detection, user-suggested translation corrections, browser extension and API integration and so on.

Despite technological advances, Google Translate still faces challenges with polysemy (multiple word meanings) and idiomatic expressions. Translation accuracy varies by language and text complexity, and outputs may contain errors[4].

2.8.3 DeepL Translator

DeepL Translator is a multilingual machine translation service based on neural networks, developed by DeepL GmbH. The service leverages the knowledge base of Linguee, another service created by the same company. DeepL offers a free translation solution that doesn't require registration, along with a premium version that includes an API for desktop and business application integration[28]. For more information, visit the official DeepL website: [DeepL Translator](https://www.deepl.com/).

DeepL uses convolutional neural networks and high-performance computing servers for translations. Its algorithm and source code are proprietary. According to tests using the BLEU algorithm, DeepL Translator's quality matches or exceeds major competing services like Google Translate, Microsoft Translator, and Facebook[8].

As of 2023, the platform supports 31 languages, enabling 552 language combinations. The service provides text translations through both a free basic version and a paid version with API access. It's available as software for Windows and macOS, and as a free add-on for Google Chrome and Mozilla Firefox[28].

In July 2024, DeepL released a large language model optimized for linguistic translation, trained on proprietary data collected over seven years and refined through input from linguistic experts. The DeepL website complies with European GDPR data protection regulations[22].

Regarding popularity, in January 2019, the site had approximately 312,000 daily unique users and 2.5 million views. DeepL ranked 150th on toptools4learning.com as personal productivity software. Thirty percent of users are concentrated in Germany, followed by France, Switzerland, Spain, and Italy[25].

Unlike Google Translate, which evolved from statistical machine translation to neural machine translation with Transformer architectures, DeepL has always used neural networks, particularly convolutional ones. The service emphasizes proprietary language models and carefully curated data collected over time. While both services offer high-quality results, DeepL distinguishes itself through its focus on precision, use of proprietary data, and dedicated language model[22].

2.9 Workflow of Translation and Retranslation

The translation and retranslation workflow is designed to assess how automated similarity metrics respond to variations introduced through multi-step language transformations. The process consists of the following steps:

1. **Translation:** The original sentence is translated from English into a target language (e.g., French) using a selected translation service (Google Translate or DeepL).
2. **Retranslation:** The translated sentence is then retranslated back into English using the same service, completing the translation-retranslation cycle.
3. **Multi-Service Variation:** The translation-retranslation process is performed separately for both Google Translate and DeepL to evaluate differences in how each

system affects similarity metrics.

4. **Paraphrased Sentence Inclusion:** Some sentences undergo an additional step where they are paraphrased using a layered paraphrasing approach before translation to observe how this affects similarity scores.
5. **Similarity Analysis:** The resulting retranslated sentences are compared against the original English sentences using multiple similarity metrics (BLEU, ROUGE, cosine similarity, etc.).
6. **Human Evaluation Alignment:** The similarity scores are compared to human ratings to analyze how well the automated metrics align with human perception of similarity.

2.10 Challenges in Multilingual Translation

Evaluating watermark robustness in a multilingual context introduces several challenges that can impact similarity measurements and watermark persistence:

- **Linguistic Variability:** Different languages exhibit significant differences in syntax, grammar, and vocabulary. These structural variations can lead to differing levels of distortion during translation and retranslation, affecting the fidelity of the original text.
- **Translation Model Differences:** Each translation service relies on distinct neural architectures and training datasets, leading to variations in translation style, choice of words, and potential biases. This introduces inconsistencies in how watermarks are preserved across different translation pipelines.
- **Semantic Drift:** Retranslation may introduce subtle changes in meaning, leading to deviations from the original intent. These semantic shifts can lower similarity scores and affect the reliability of metrics such as cosine similarity and Z-scores.

Despite these challenges, the translation-retranslation process remains a valuable method for stress-testing watermark robustness in real-world multilingual scenarios.

Chapter 3

Results and Analysis

3.1 Introduction

This chapter presents an in-depth analysis of the methodology, data processing steps, and results obtained from evaluating the performance of similarity metrics in assessing text similarity across multiple languages and translation services. The study was designed to systematically investigate how different transformations affect similarity scores and their alignment with human perception. The analysis goes through several stages, starting with dataset preparation, followed by layered paraphrasing, translation and retranslation, metric evaluation, and finally, human evaluation.

3.2 Dataset Preparation

To ensure a robust experimental setup, the study utilized the first 100 sentences extracted from the Microsoft Research Paraphrase Corpus (MRPC). This dataset is widely used in natural language processing (NLP) research and consists of sentence pairs labelled as semantically equivalent or non-equivalent. The selected subset was chosen because it includes a diverse range of linguistic structures and semantic variations, serving as a strong foundation for subsequent transformations.

3.3 Layered Paraphrasing Approach

A layered paraphrasing approach was implemented to introduce subtle variations in the text while preserving semantic meaning. This approach is crucial for testing how well similarity metrics capture meaning-preserving transformations. The methodology consists of three primary transformation layers: grammar transformations, contextual paraphrasing, and BERT-based contextual replacements. In order to assess the sensitivity of similarity metrics to varying degrees of textual modification, two separate configurations were tested for both the T5-based paraphrasing model and the BERT-based word replacement approach.

3.3.1 Grammar Transformations

The first layer involved making minor grammatical modifications to the text. These transformations were designed to test whether minor syntactic changes, such as verb tense alterations, auxiliary modifications, or contractions, would impact similarity scores. Examples of transformations include:

- "She *is* going to the market." → "She *was* going to the market."
- "They *have* seen the movie." → "They *had* seen the movie."
- "He *will* call you soon." → "He *would* call you soon."

Such modifications were applied systematically across all selected sentences, ensuring that no additional semantic meaning was introduced. This process helped assess the robustness of similarity metrics to minor variations in wording.

3.3.2 Contextual Paraphrasing

The second transformation layer utilized a T5-based paraphrasing model to generate alternative versions of the original text. This step introduced additional variability while maintaining the core meaning. The goal was to observe how similarity metrics responded to semantically equivalent but structurally different sentences.

Two distinct configurations of the T5-based paraphraser were employed:

1. A **conservative paraphrasing** approach, in which the model was configured with

a lower temperature (1.2) and a high top-p value (0.90). This setting generated paraphrases that were closer to the original sentence while still introducing minor lexical variations. The output was deterministic, meaning that each input sentence produced only one paraphrased alternative.

2. A **more aggressive paraphrasing** approach, where the model was configured with a higher temperature (1.8) and a lower top-p value (0.8). Additionally, multiple paraphrases were generated per sentence, and the least similar alternative (as determined by cosine similarity between sentence embeddings) was selected. This ensured that the paraphrased sentences exhibited greater lexical and syntactic divergence from the original, simulating stronger perturbations.

By employing these two configurations, the study aimed to determine whether similarity metrics remained robust across both minor and more pronounced paraphrastic modifications. Metrics that performed well under both settings could be considered more reliable in detecting meaning-preserving transformations.

3.3.3 BERT-Based Contextual Replacements

The final transformation layer involved replacing specific words within the text with semantically similar alternatives using a BERT-based masked language model. This transformation step introduced variations at the lexical level without significantly altering the sentence structure. The purpose of this step was to evaluate whether similarity metrics could detect meaning equivalence when minor lexical substitutions were applied.

As with the paraphrasing step, two different configurations were tested:

1. A **higher similarity threshold** setting (0.75) that replaced words only if the substitute had a high cosine similarity with the original. This ensured that the replacements closely aligned with the original meaning, preserving the sentence’s overall structure while making small lexical changes.
2. A **lower similarity threshold** setting (0.5), which allowed for more diverse replacements by selecting words that were less semantically similar to the original. Additionally, this configuration prioritized selecting the least similar valid replacement among the top-k suggestions. This approach aimed to simulate stronger per-

turbations and test whether similarity metrics could still recognize the underlying semantic meaning.

By introducing two levels of word replacement severity, this experiment provided a deeper understanding of how different similarity metrics respond to varying degrees of lexical modification.

3.3.4 Summary of Transformation Strategies

The combination of grammar transformations, paraphrasing strategies, and BERT-based replacements enabled a comprehensive evaluation of text similarity metrics under controlled transformations. By comparing the performance of metrics across these variations, the study aimed to answer the following key questions:

- Do similarity metrics remain stable across different degrees of paraphrasing?
- How well do these metrics handle minor grammatical and lexical changes?
- Are embedding-based similarity measures (such as cosine similarity) more robust than traditional n-gram metrics (e.g., BLEU, ROUGE)?
- Does a more aggressive perturbation strategy lead to significantly lower similarity scores, even when meaning is preserved?

The results obtained from these transformations provide valuable insights into the robustness of similarity metrics and their alignment with human perception of textual similarity, which will be further discussed in the subsequent chapters.

3.4 Translation and Re-translation Process

To assess how translation services impact similarity metrics, the paraphrased sentences underwent a translation-retranslation process using two popular translation services: Google Translate and DeepL. This step aimed to introduce natural variations that occur during machine translation while preserving the overall meaning of the text.

3.4.1 Choice of Languages

A total of 10 target languages were selected for translation, covering a mix of high-resource and medium-resource languages:

- **High-resource languages:** French, Spanish, German, Italian, Russian, Portuguese
- **Medium-resource languages:** Swedish, Japanese, Dutch, Turkish

The selection ensured a fair linguistic diversity, allowing for a comprehensive evaluation of how different languages affect text transformations and similarity scores. Other languages could have been taken in consideration as well, but due to computational and DeepL’s API usage limits it was decided not to.

3.4.2 Translation and Re-Translation

Each sentence was translated from English into a target language and then retranslated back into English using both Google Translate and DeepL. This approach helped measure the stability of similarity metrics under real-world conditions where minor variations in translations are common.

Observations from the translation-retranslation process revealed that some translations introduced idiomatic expressions, altering the structure while preserving meaning. Certain languages (e.g., Japanese) tended to produce more structural changes due to syntactic differences, and back-translations often introduced slight rewordings that impacted similarity scores.

These insights played a crucial role in evaluating how similarity metrics respond to real-world linguistic variations.

3.5 Selection of Sentence Pairs for Human Evaluation

Following the transformation and translation steps, a subset of 94 sentence pairs was selected for human evaluation. The selection process aimed to capture a wide range of similarity scores across different metrics, ensuring a meaningful comparison between

computational and human judgments.

3.5.1 Selection Criteria

The selection process was guided by specific criteria designed to ensure the dataset's diversity and relevance. The criteria included:

- **Diversity in Structure:** The selected sentences were chosen to represent a variety of structures, including simple, compound, and complex sentences. This ensured that the dataset encompassed different levels of linguistic complexity, providing a more nuanced evaluation.
- **Semantic Coverage:** The dataset included a mix of sentence pairs with high, moderate, and low similarity scores. This range allowed for a comprehensive assessment of how well the similarity metrics performed across different levels of semantic equivalence.

To facilitate this selection process, a Python script was employed. The script utilized a function to identify and select sentence pairs based on predefined criteria related to their similarity metrics. The function `select_interesting_pairs` was designed to filter and randomly select pairs from the dataset that met specific conditions. These conditions included combinations of high and low scores across different metrics, such as Cosine Similarity, BLEU, ROUGE-L, and METEOR.

The script first defined several criteria to identify "interesting" pairs, such as those with low Cosine Similarity but high METEOR scores, or high Cosine Similarity but low BLEU scores. These criteria were chosen to highlight pairs that demonstrated notable differences or alignments in their similarity metrics, providing a rich dataset for analysis. The specific criteria used in the script were as follows:

- Low Cosine Similarity but high METEOR scores.
- High Cosine Similarity but low BLEU scores.
- High BLEU but low METEOR scores.
- High ROUGE-L but low Cosine Similarity scores.
- Large differences between ROUGE-1 and ROUGE-L scores.

- Large differences between ROUGE-2 and ROUGE-L scores.
- Combinations of low Cosine Similarity, high BLEU, and low METEOR scores.

The script then combined the results from these criteria, removed duplicates, and randomly selected the desired number of pairs. If the number of pairs meeting the criteria was insufficient, the script randomly selected additional pairs from the entire dataset to ensure the required number of pairs was met.

This systematic approach ensured that the selected sentence pairs were not only diverse in structure and semantic coverage but also representative of various similarity metric combinations, providing a robust foundation for the human evaluation process.

3.6 Human Evaluation

A survey was conducted using Google Forms to assess how well computational similarity metrics align with human perception. Participants were presented with pairs of sentences and asked to rate their similarity on a predefined scale.

3.6.1 Survey Design and Procedure

Six participants evaluated a set of 94 sentence pairs selected based on diverse similarity metric values. The sentences were displayed without any transformation labels to prevent bias. Participants rated similarity based on meaning rather than surface-level structure, using the following scale:

- **Completely different** (The two sentences have no meaning in common.) [Value = 0.0]
- **Slightly similar** (There are some shared words, but the meaning is mostly different.) [Value = 0.25]
- **Moderately similar** (The meaning is somewhat preserved, but there are significant changes.) [Value = 0.5]
- **Very similar** (The meaning is mostly the same with minor wording differences.) [Value = 0.75]

- **Almost identical** (The two sentences convey exactly the same meaning.) [Value = 1.0]

Next to the scale there have been reported the corresponding values of the selectable options used for comparing them with the similarity metrics.

Each survey question followed a standardized format. A sample question was:

How would you rate this pair of sentences?

- "Meanwhile, the global death toll approached 770 with more than 8,300 people sickened since the severe acute respiratory syndrome virus first appeared in southern China in November."
- "At the same time, the international death toll has exceeded 500 after more than 100 people have been hospitalized since the flu illness began in mainland Korea on July 1."

3.6.2 Handling Annotator Disagreement

Since multiple annotators rated each sentence pair, discrepancies in responses were observed. To derive a final human similarity rating for each sentence pair, we computed the **mean** of all annotator scores. This method ensures that extreme judgments do not disproportionately affect the results while maintaining a balanced representation of human perception.

For example, in the sentence pair shown above:

- Two annotators rated it as **Completely different** (0.0).
- One annotator rated it as **Slightly similar** (0.25).
- Three annotators rated it as **Moderately similar** (0.5).

The final human similarity score was calculated as:

$$\text{Human Rating} = \frac{(2 \times 0.0) + (1 \times 0.25) + (3 \times 0.5)}{2 + 1 + 3} \approx 0.3 \quad (3.1)$$

This averaging method was chosen over other aggregation methods, such as the median or mode, because it captures the overall tendency of human perception while smoothing

out outlier ratings.

3.6.3 Comparison with Computational Metrics

To further investigate differences between human and computational judgments, we compared the derived human ratings with similarity scores obtained from different computational metrics.

For the same sentence pair, the computed similarity scores were:

- **Cosine Similarity:** 0.8173
- **BLEU:** 0.0000
- **ROUGE- L_{f1} :** 0.2963
- **METEOR:** 0.3270

The high cosine similarity (0.8173) indicates that embedding-based similarity models detected strong similarity, whereas BLEU (0.0) failed to capture any resemblance due to its reliance on exact n-gram matches. The ROUGE- L_{f1} (0.2963) and METEOR (0.3270) scores were closer to the human rating of 0.3, but still showed minor deviations.

This example illustrates that certain metrics (such as BLEU) struggle with paraphrased and semantically similar text, while others (such as METEOR) more closely approximate human judgments.

3.6.4 Preliminary Observations

Initial analysis of human ratings revealed:

- High agreement on clear paraphrases but greater variability on translated and re-translated texts.
- A tendency for human raters to assign lower similarity scores to sentences with structural changes, even if meaning was preserved.
- Instances where similarity metrics overestimated similarity compared to human judgment, particularly for back-translated sentences.

These insights provided a strong foundation for comparing computational metrics against human evaluations in the next phase of analysis.

Now let’s see a detailed analysis of the effectiveness of computational similarity metrics in assessing textual similarity compared to human evaluations. We investigate widely used metrics, including BLEU, ROUGE, METEOR, cosine similarity, and BERT-based embeddings, to determine their reliability in capturing semantic equivalence. We also explore correlation analysis and ROC curve evaluations to assess the discriminative power of these metrics.

3.7 Correlation Analysis

To analyze how well computational similarity metrics align with human perception, we computed the Pearson correlation coefficient between the aggregated human ratings and the similarity scores obtained from different computational metrics.

The final human similarity scores, obtained as described in the previous section, were used as the ground truth in correlation analysis. Since annotators sometimes provided different ratings for the same sentence pair, we computed the mean of all ratings to create a single aggregated score per sentence pair. These averaged scores were then used to compute Pearson correlation coefficients with computational similarity metrics. We then computed the correlation of these scores with four computational metrics: Cosine Similarity, BLEU, ROUGE- L_{f1} , and METEOR. Table 3.1 presents the correlation coefficients and their statistical significance (p-values).

Metric	Correlation	p-value
Cosine Similarity	0.4508	5.13×10^{-6}
BLEU	0.6471	1.83×10^{-12}
ROUGE- L_{f1}	0.6888	1.68×10^{-14}
METEOR	0.8007	3.45×10^{-22}

Table 3.1: Correlation of similarity metrics with human judgments.

The results indicate that METEOR exhibits the highest correlation (0.8007) with human judgments, followed by ROUGE- L_{f1} (0.6888) and BLEU (0.6471). The Cosine

Similarity metric, though showing a positive correlation, has the lowest value (0.4508), suggesting a weaker alignment with human assessments. The extremely low p-values across all metrics indicate that these correlations are statistically significant, reinforcing the reliability of these findings.

The high correlation of METEOR suggests that it is particularly effective in capturing semantic equivalence, even in the presence of structural variations introduced during paraphrasing and translation. ROUGE- L_{f1} also shows strong performance, indicating its reliability in evaluating text similarity, though it is less sensitive to semantic nuances compared to METEOR. BLEU, while showing moderate correlation, tends to overestimate similarity for texts with high lexical overlap but differing semantics, which may explain its lower performance relative to METEOR and ROUGE- L_{f1} .

3.8 Precision, Recall, and F1-score Analysis

To further analyze the performance of these metrics, we computed precision, recall, and F1-score for each. Table 3.2 presents the results:

Metric	Precision	Recall	F1-score
Cosine Similarity	0.7204	1.0000	0.8375
BLEU	0.9545	0.3134	0.4719
ROUGE- L_{f1}	0.9535	0.6119	0.7455
METEOR	0.9643	0.8060	0.8780

Table 3.2: Precision, recall, and F1-score for different similarity metrics.

METEOR again outperforms other metrics with an F1-score of 0.8780, indicating strong precision and recall balance. This suggests that METEOR effectively captures both the presence and absence of semantic similarity, making it a robust choice for evaluating text similarity. BLEU, despite high precision (0.9545), suffers from poor recall (0.3134), leading to a relatively low F1-score (0.4719). This indicates that BLEU is more effective at identifying true positives but struggles with false negatives, likely due to its focus on lexical rather than semantic similarity.

ROUGE- L_{f1} provides a more balanced performance with an F1-score of 0.7455, re-

flecting its ability to capture both precision and recall effectively. However, its performance is still inferior to METEOR, particularly in recall, suggesting that while ROUGE- L_{f1} is reliable, it may miss some semantically similar pairs that METEOR can detect.

3.9 Correlation of Precision, Recall, and F1-score

We further analyzed the correlation between metric performance (precision, recall, and F1-score) and the human labels. The results are shown in Table 3.3.

Metric	Correlation	p-value
Cosine Similarity (Precision/Recall/F1)	0.8501	2.29×10^{-27}
BLEU (Precision/Recall/F1)	0.5245	5.76×10^{-8}
ROUGE- L_{f1} (Precision/Recall/F1)	0.6253	1.61×10^{-11}
METEOR (Precision/Recall/F1)	0.7787	2.51×10^{-20}

Table 3.3: Correlation of precision, recall, and F1-score with human judgments.

These results reinforce METEOR’s reliability as it demonstrates the highest correlation (0.7787). This indicates that METEOR not only aligns well with human judgments in terms of overall correlation but also performs consistently across precision, recall, and F1-score. In contrast, BLEU has the lowest correlation (0.5245), further supporting its weaker recall performance and suggesting that it is less reliable for capturing the nuances of semantic similarity.

3.10 Optimal Thresholds for Classification

To determine optimal thresholds for classification using these metrics, we identified the values that maximize performance by maximizing the difference between the true positive rates and the false positive rates, essentially finding the point on the ROC curve closest to the top-left corner (ideal performance). The thresholds are summarized in Table 3.4.

These thresholds suggest that METEOR and ROUGE- L_{f1} require higher cutoff values to distinguish between similar and dissimilar texts, whereas BLEU has the lowest threshold, likely due to its limited sensitivity to minor paraphrasing. The relatively

Metric	Optimal Threshold
Similarity Score	0.8199
BLEU	0.136
ROUGE-L _{f1}	0.3571
METEOR	0.5173

Table 3.4: Optimal thresholds for classification.

high threshold for the Similarity Score indicates that it may be overly conservative in classifying texts as similar, potentially missing some true positives.

Overall, METEOR emerges as the most reliable similarity metric, offering a strong balance of correlation, F1-score, and classification performance. Its ability to capture both lexical and semantic similarity makes it a valuable tool for evaluating text similarity in diverse linguistic contexts.

3.11 Analysis of Similarity Metrics

In this section, we analyze the behaviour and effectiveness of various similarity metrics used to evaluate the textual similarity between original and perturbed texts. The goal is once again to assess how well computational metrics align with human perception and determine the robustness of similarity scores under different transformation processes. We have used multiple visualizations and they are:

3.11.1 Precision-Recall Curve

The precision-recall curve (Figure 3.1) illustrates the trade-off between precision and recall for different similarity thresholds. This is particularly relevant in scenarios where the dataset is imbalanced, as is often the case in textual similarity detection.

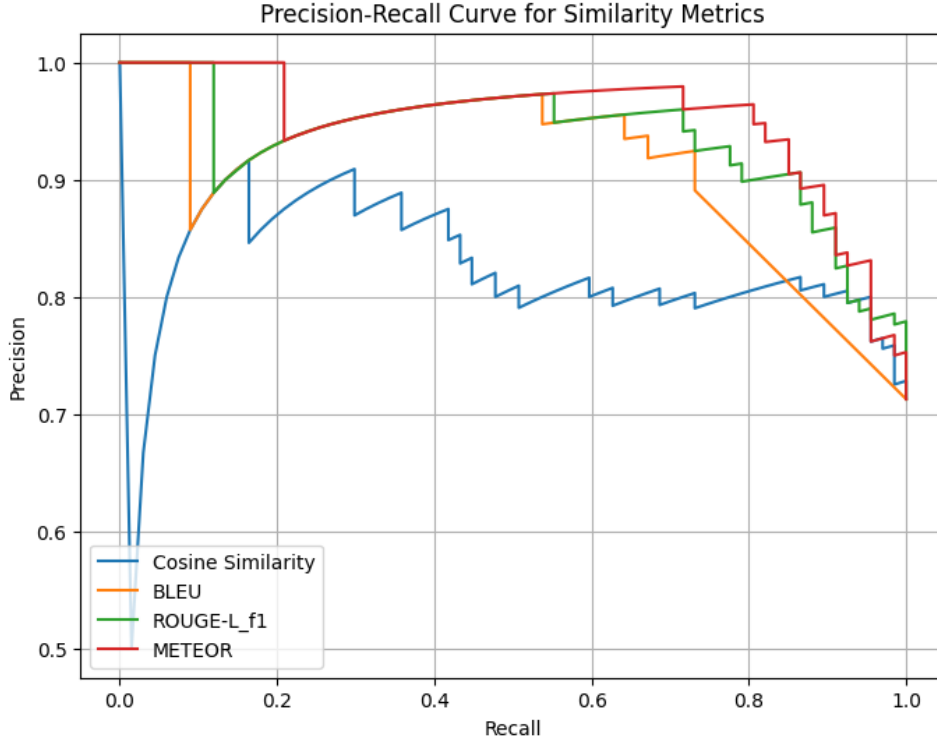


Figure 3.1: Precision-recall curve for different similarity metrics.

The x-axis of the PR curve represents recall, which indicates how many actual plagiarism cases were correctly detected by the metric. The y-axis represents precision, showing how many of the detected cases were truly instances of plagiarism. A higher area under the curve (AUC-PR) signifies better performance in detecting plagiarism, as it reflects a more effective balance between precision and recall.

Examining the performance of each metric, we observe that METEOR, represented by the red curve, demonstrates the best overall performance. It maintains high precision, exceeding 90%, across a wide range of recall values. This stability indicates that METEOR is highly effective in correctly identifying plagiarism cases while minimizing false positives. At low recall values, METEOR achieves perfect precision, and as recall increases, it continues to maintain high precision. This consistency makes METEOR the most reliable metric for plagiarism detection among those evaluated.

ROUGE-L F1, depicted by the green curve, performs slightly worse than METEOR but still maintains a commendable level of precision across most recall values. While it does not match METEOR's stability, ROUGE-L F1 remains a strong contender, particularly in scenarios where a balance between precision and recall is crucial. Its performance

suggests it is a good alternative to METEOR, though not as robust in all conditions.

In contrast, BLEU, shown by the orange curve, performs poorly in comparison to METEOR and ROUGE-L F1. Its precision drops rapidly as recall increases, indicating that it struggles to maintain accuracy as more plagiarism cases are detected. This decline in performance suggests that BLEU is not a reliable metric for plagiarism detection, as it fails to capture the nuances of text similarity effectively.

The blue curve, representing the Cosine Similarity, shows more variable precision, fluctuating significantly as recall increases. While it starts with lower precision, it improves as recall rises, indicating some level of effectiveness. However, its instability makes it less reliable than METEOR and ROUGE-L F1. The inconsistency in its performance suggests that while the Cosine Similarity can detect plagiarism to some extent, its unreliable nature makes it less suitable for practical applications.

The implications of these observations are significant for your work. METEOR emerges as the best metric for plagiarism detection, offering the most balanced performance between precision and recall. Its ability to maintain high precision across varying recall values makes it an ideal choice for applications requiring accurate plagiarism detection. ROUGE-L F1 also proves to be a strong alternative, providing reliable performance, though slightly less consistent than METEOR.

BLEU's poor performance indicates it is not well-suited for plagiarism detection, as it fails to effectively distinguish between plagiarized and non-plagiarized texts. The Similarity Score, while showing some potential, is too unstable to be relied upon for consistent detection.

3.11.2 Receiver Operating Characteristic Curve

The ROC curve (Figure 3.2) provides insight into the discriminatory power of similarity metrics by plotting the true positive rate (sensitivity) against the false positive rate (1 - specificity).

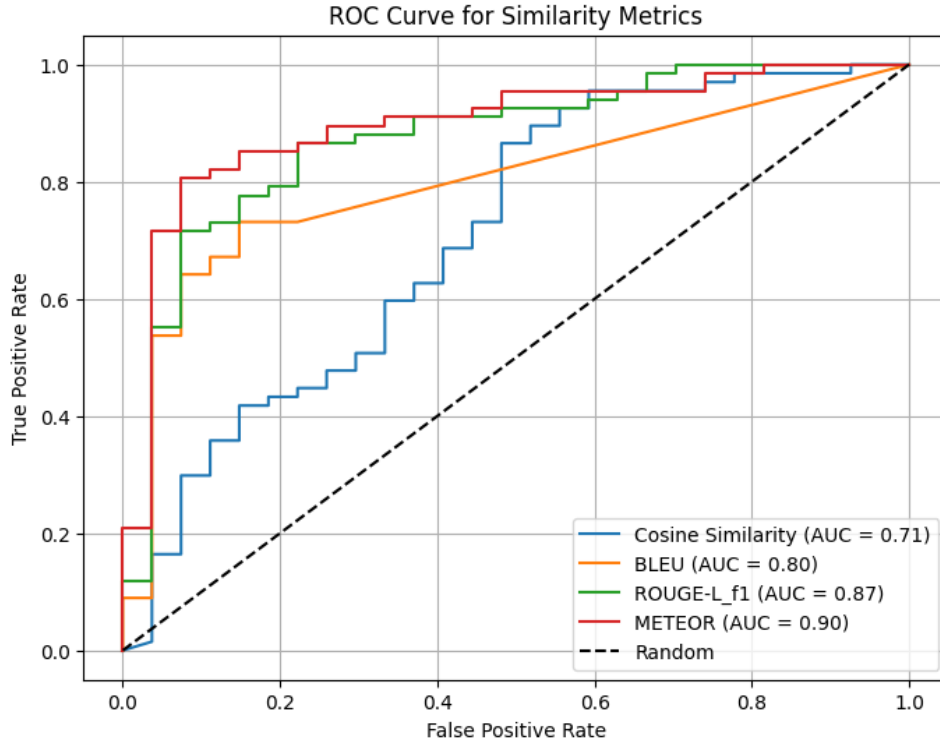


Figure 3.2: ROC curve for various similarity metrics.

The AUC, or Area Under the ROC Curve, is a critical metric for evaluating the performance of a classifier, particularly in distinguishing between positive and negative instances. In the context of your study, the positive instances are plagiarized texts, while the negative instances are non-plagiarized texts. The AUC represents the probability that a randomly chosen plagiarized text will be ranked higher by the model than a randomly chosen non-plagiarized text.

Understanding the AUC scale is essential for interpreting the results. An AUC of 1.0 signifies a perfect classifier, one that can flawlessly distinguish between plagiarized and non-plagiarized texts. On the other hand, an AUC of 0.5 indicates a classifier that performs no better than random chance, essentially equivalent to flipping a coin. Any AUC value greater than 0.5 suggests that the classifier is better than random chance, with higher values indicating superior performance.

Let's see what the AUC values you obtained tell us about each metric:

Starting with Cosine Similarity, which has an AUC of 0.71, we see that this metric performs better than random chance in distinguishing between plagiarized and non-

plagiarized text. However, its performance is moderate, indicating there is room for improvement. Cosine Similarity captures some aspects of text similarity but may not be as effective in discerning the nuances that separate plagiarized from non-plagiarized content.

BLEU, with an AUC of 0.80, shows better performance than Cosine Similarity. This higher AUC suggests that BLEU has a greater probability of correctly ranking plagiarized text higher than non-plagiarized text.

ROUGE-L F1 and METEOR both demonstrate exceptional performance, each with an AUC of 0.90. These high AUC values indicate that both metrics are highly effective in distinguishing between plagiarized and non-plagiarized texts. ROUGE-L F1's strength comes from its focus on recall and the longest common subsequence, making it sensitive to the retention of key phrases and structures in the text. METEOR, on the other hand, benefits from its ability to account for synonyms and stemming, providing a more nuanced measure of text similarity.

Overall, ROUGE-L F1 and METEOR emerge as the most effective metrics for your plagiarism detection task, based on their high AUC values. Their superior performance suggests they are better equipped to handle the complexities of text similarity and plagiarism detection compared to Cosine Similarity and BLEU. While BLEU performs better than Cosine Similarity, it does not reach the level of effectiveness demonstrated by ROUGE-L F1 or METEOR.

Cosine Similarity, with the lowest AUC among the metrics, suggests it might not be the most suitable metric for plagiarism detection in this context. Its moderate performance indicates that while it captures some aspects of text similarity, it may not be as sensitive to the specific characteristics that define plagiarism.

It's important to consider that while AUC provides an overall measure of performance, selecting a specific threshold on the ROC curve is necessary for practical application. The optimal threshold will depend on the specific needs of the task and the desired balance between false positives and false negatives. Additionally, these results are based on our specific limited dataset and experimental setup, and performance may vary with different datasets or types of text.

3.11.3 Scatter Plots of Similarity Scores

To further explore the distribution of similarity scores, we present scatter plots comparing different metrics against human similarity judgments (Figure 3.3).

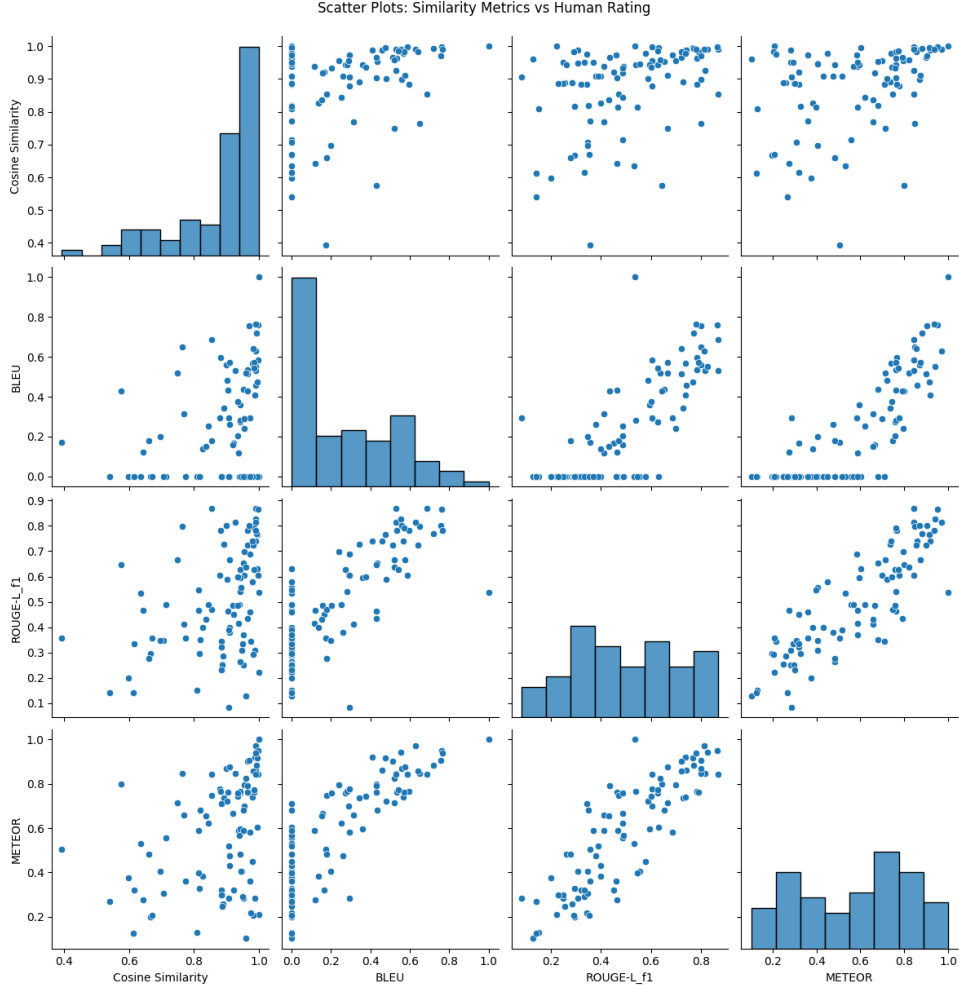


Figure 3.3: Scatter plot of computed similarity scores vs. human similarity judgments.

The plot is organized as a 4x4 grid, where the diagonal elements, stretching from the top-left to the bottom-right, are histograms showing the distribution of each metric's scores. These histograms provide a visual summary of how each metric's scores are spread across different rating bins. The off-diagonal elements are scatter plots that compare pairs of metrics, allowing us to see how the scores of one metric relate to the scores of another for the same sentence pairs.

Starting with the diagonal histograms, we observe the distribution of scores for each metric. For Cosine Similarity, the histogram shows that most scores are clustered around

higher values, indicating that many sentence pairs have high similarity according to this metric. This suggests that Cosine Similarity tends to assign high similarity scores to a majority of the sentence pairs.

In contrast, the BLEU scores are more spread out, with a significant number of low scores. This distribution suggests that BLEU is more sensitive to variations in text and tends to assign lower similarity scores more frequently. This sensitivity makes BLEU useful for detecting subtle differences in text but may also lead to lower scores for texts that are semantically similar but lexically diverse.

The ROUGE-L F1 scores show a relatively uniform distribution, indicating a balanced sensitivity to text variations. This metric seems to capture a middle ground between the high sensitivity of BLEU and the more lenient scoring of Cosine Similarity and METEOR.

METEOR, like Cosine Similarity, has scores clustered around higher values, suggesting that it generally assigns high similarity scores to many sentence pairs. This indicates that METEOR is effective in capturing semantic similarity, even when there are variations in the text.

Moving to the off-diagonal scatter plots, each plot compares two different metrics. For example, the scatter plot comparing Cosine Similarity on the x-axis and BLEU on the y-axis shows a wide spread of points. This indicates that Cosine Similarity and BLEU do not have a strong linear relationship, suggesting that these metrics capture different aspects of text similarity. Cosine Similarity tends to assign higher scores compared to BLEU, which is evident from the clustering of points at higher Cosine Similarity values but lower BLEU scores.

The scatter plot comparing Cosine Similarity and ROUGE-L F1 shows a more concentrated cluster of points, indicating a stronger relationship between these two metrics. Both metrics tend to assign higher scores to similar sentence pairs, suggesting they capture similar aspects of text similarity.

Similarly, the scatter plot comparing Cosine Similarity and METEOR shows a little bit of correlation. This indicates that both metrics capture similar aspects of text similarity and assign high scores to similar sentence pairs.

The scatter plot comparing BLEU and ROUGE-L F1 shows a moderate spread of

points, indicating a weaker relationship between these two metrics. BLEU tends to assign lower scores compared to ROUGE-L F1, suggesting that BLEU is more sensitive to variations in text.

The scatter plot comparing BLEU and METEOR also shows a wide spread of points, indicating a weaker relationship between these two metrics. BLEU tends to assign lower scores compared to METEOR, suggesting that METEOR is more robust to variations in text.

Finally, the scatter plot comparing ROUGE-L F1 and METEOR shows a strong correlation, with points clustered along the diagonal. This indicates that both metrics capture similar aspects of text similarity and assign high scores to similar sentence pairs.

The implications of these observations are significant. The strong correlations between Cosine Similarity, ROUGE-L F1, and METEOR suggest that these metrics capture similar aspects of text similarity. This alignment indicates that these metrics are reliable for evaluating text similarity in various contexts. On the other hand, BLEU shows weaker correlations with other metrics, suggesting it captures different aspects of text similarity. Its sensitivity to variations in text makes it useful for specific applications but less reliable for general text similarity evaluation.

The alignment of these metrics with human judgments, as indicated by the clustering of high scores, suggests that they are effective in capturing semantic similarity as perceived by humans. This makes them valuable tools for applications requiring accurate text similarity evaluation.

3.11.4 Correlation Matrix

This plot (Figure 3.4) is a correlation matrix heatmap that visually represents the correlation coefficients between different similarity metrics and human ratings. Each cell in the matrix shows the correlation value between two metrics or between a metric and human ratings. The color intensity and scale on the right indicate the strength and direction of the correlation.

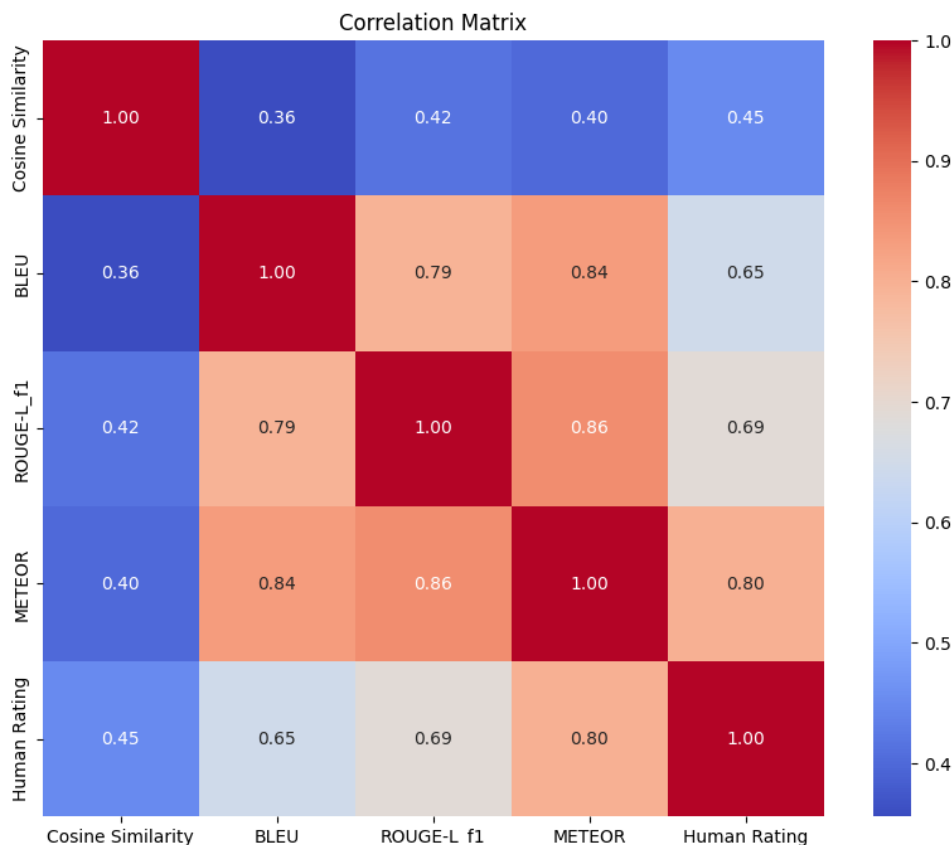


Figure 3.4: Correlation map of the correlation between the metrics and human ratings.

The structure of the plot is organized with both the x-axis and y-axis listing the metrics and human ratings being compared, which include Cosine Similarity, BLEU, ROUGE-L F1, METEOR, and Human Rating. The diagonal cells, stretching from the top-left to the bottom-right, represent the correlation of each metric with itself, which is always 1.0, as a metric is perfectly correlated with itself. The off-diagonal cells, on the other hand, display the correlation coefficients between different pairs of metrics or between a metric and human ratings.

The color scale on the right of the plot helps interpret the strength of the correlations. Dark red indicates a strong positive correlation, close to 1.0, while lighter colors represent moderate correlations. Although dark blue would typically indicate a strong negative correlation, this plot only shows positive values, as all the correlations here are non-negative.

Examining the correlations, we see that Cosine Similarity has a correlation of 0.45 with Human Rating, indicating a moderate positive relationship. This suggests that Co-

sine Similarity captures some aspects of human-perceived similarity but is not as strongly aligned as other metrics. When compared to other metrics, Cosine Similarity shows moderate correlations with BLEU (0.36) and ROUGE-L F1 (0.42), suggesting that while there is some overlap, Cosine Similarity captures different aspects of text similarity.

BLEU, with a correlation of 0.65 with Human Rating, shows a stronger relationship with human judgments compared to Cosine Similarity. This indicates that BLEU captures aspects of text similarity that align more closely with human perception. BLEU also shows strong correlations with ROUGE-L F1 (0.79) and METEOR (0.84), indicating that it shares commonalities with these metrics in evaluating text similarity.

ROUGE-L F1, with a correlation of 0.69 with Human Rating, suggests a strong alignment with human judgments. This metric effectively captures text similarity as perceived by humans. Its high correlations with BLEU (0.79) and METEOR (0.86) indicate that ROUGE-L F1 is closely related to these metrics in its evaluation approach.

METEOR exhibits the highest correlation with Human Rating at 0.80, indicating that it aligns most closely with human judgments of text similarity. Its very strong correlations with BLEU (0.84) and ROUGE-L F1 (0.86) suggest that METEOR integrates aspects of both metrics effectively.

The correlations with human ratings provide insight into how well each metric aligns with human perception. METEOR shows the strongest alignment, followed by ROUGE-L F1 and BLEU, with Cosine Similarity being the least aligned. This suggests that METEOR is the most effective metric for capturing human-perceived text similarity, while ROUGE-L F1 and BLEU are also reliable. The moderate correlations between Cosine Similarity and other metrics indicate that it captures different aspects of text similarity, providing complementary insights when used alongside other metrics.

3.11.5 Box Plots

The box plots provide a detailed visualization of how different similarity metrics align with human ratings across various levels of perceived similarity. Each plot represents the distribution of a specific metric's scores across different rating bins, which are categorized as Low, Medium-Low, Medium-High, and High, reflecting varying degrees of human-

perceived similarity. The metrics analyzed include Cosine Similarity, BLEU, ROUGE-L F1, and METEOR.

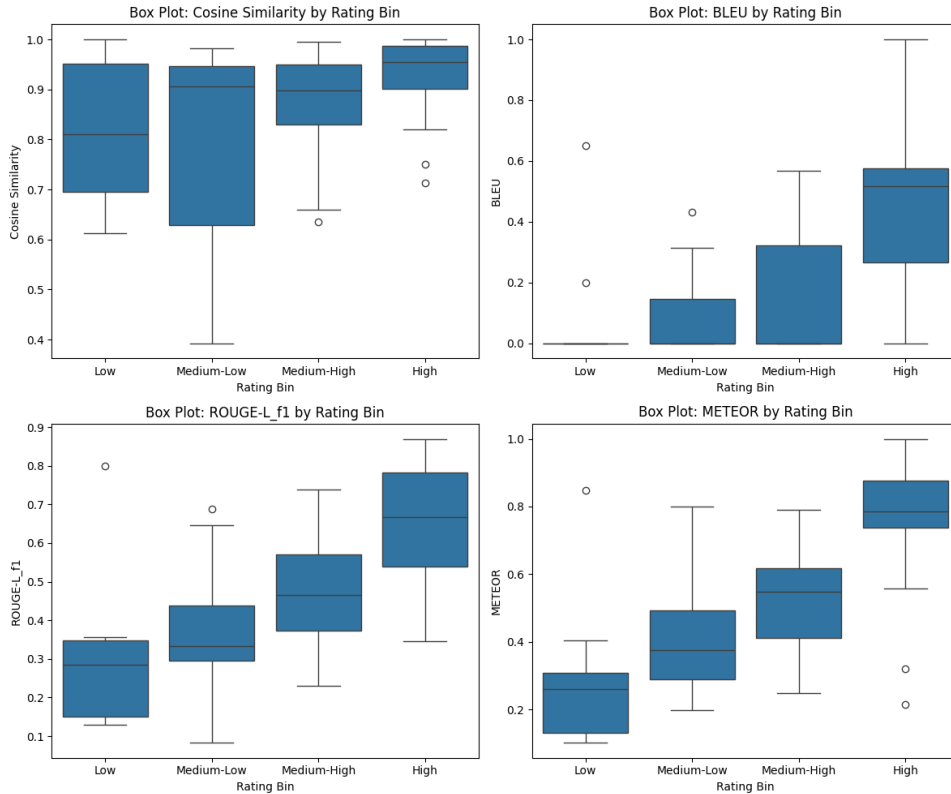


Figure 3.5: Box plot of how different similarity metrics align with human ratings.

Starting with the box plot for Cosine Similarity, we observe that the distribution of scores varies significantly across the rating bins. In the Low rating bin, cosine similarity scores are generally lower, with a median around 0.75. There is some variability, but most scores remain below 0.9, indicating that the metric assigns lower similarity to sentence pairs that humans also rate as less similar. As we move to the Medium-Low bin, the scores become more spread out, with a median around 0.85, and fewer outliers are present compared to the Low bin. This suggests that cosine similarity begins to capture more nuanced differences in text similarity as human ratings increase.

In the Medium-High bin, cosine similarity scores are higher, with a median close to 0.9, and the variability is reduced. Most scores are above 0.8, indicating a stronger alignment with higher human ratings. Finally, in the High rating bin, the scores are consistently high, with a median close to 1.0 and minimal variability. This strong agreement with high human ratings suggests that cosine similarity effectively distinguishes

between different levels of human-perceived similarity, with higher scores aligning with higher human ratings.

Moving on to the BLEU metric, the box plot reveals a different pattern. In the Low rating bin, BLEU scores are very low, with a median close to 0. This indicates that BLEU assigns low similarity to sentence pairs that humans rate as low in similarity. As we progress to the Medium-Low bin, the scores increase slightly but remain low, with a median around 0.1. The variability is more pronounced compared to the Low bin, suggesting that BLEU struggles to consistently evaluate similarity at this level. In the Medium-High bin, the scores increase further, with a median around 0.3, but the variability remains high, indicating some inconsistency in BLEU’s evaluation. Finally, in the High rating bin, the scores are higher, with a median around 0.5, but the variability is still significant. This suggests that while BLEU shows a positive trend with increasing human ratings, it exhibits more variability and lower overall scores, indicating it may not capture nuanced similarities as effectively as other metrics.

The ROUGE-L F1 box plot presents a more balanced distribution of scores across the rating bins. In the Low bin, ROUGE-L F1 scores are low, with a median around 0.3, and there is some variability. As we move to the Medium-Low bin, the scores increase, with a median around 0.5, and the variability is moderate. In the Medium-High bin, the scores increase further, with a median around 0.65, and the variability is reduced compared to the lower bins. Finally, in the High rating bin, the scores are consistently high, with a median close to 0.8 and minimal variability. This strong agreement with high human ratings suggests that ROUGE-L F1 effectively captures increasing levels of human-perceived similarity, with higher scores aligning with higher human ratings.

Lastly, the METEOR box plot shows a pattern similar to that of ROUGE-L F1. In the Low rating bin, METEOR scores are low, with a median of around 0.3, and there is some variability. As we progress to the Medium-Low bin, the scores increase, with a median of around 0.5, and the variability is moderate. In the Medium-High bin, the scores increase further, with a median around 0.65, and the variability is reduced. Finally, in the High rating bin, the scores are consistently high, with a median close to 0.9 and minimal variability. This strong agreement with high human ratings suggests that METEOR effectively distinguishes between different levels of human-perceived similarity,

with higher scores aligning with higher human ratings.

3.11.6 Mean Values for Languages and Metrics

This heatmap (Figure 3.6) provides a visual representation of the mean similarity scores for different languages across various metrics, including Cosine Similarity, BLEU, ROUGE-L, and METEOR. Each row in the heatmap corresponds to a specific language, while each column represents one of the similarity metrics. The colour intensity of each cell reflects the magnitude of the mean similarity score, with darker shades indicating higher scores.

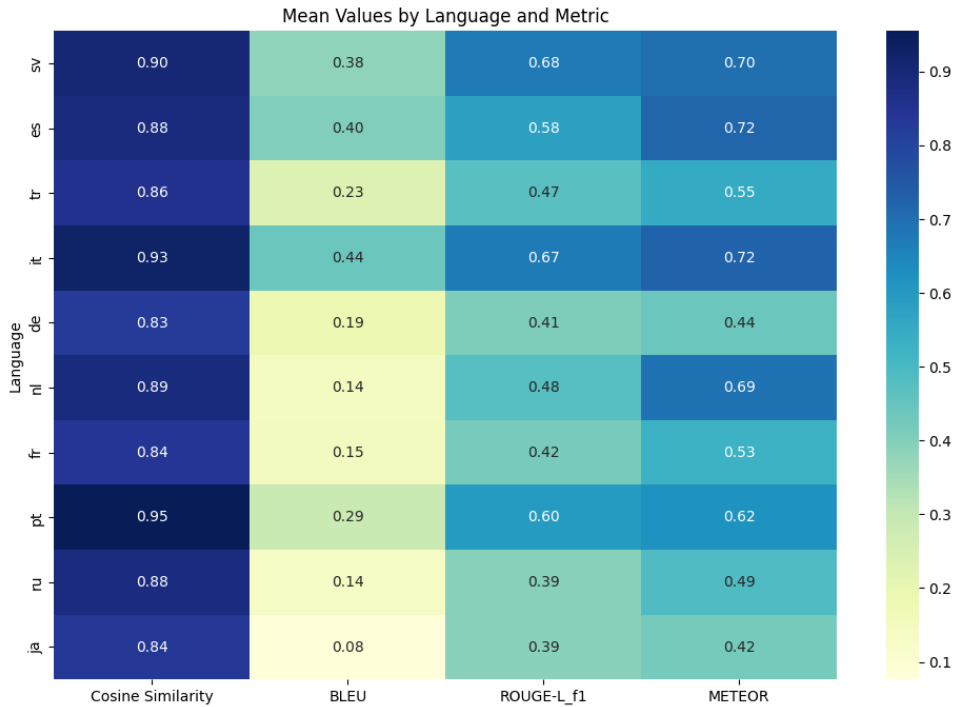


Figure 3.6: Mean similarity scores for different languages across the metrics.

The key observations that we can extract from this plot are:

1. **Cosine Similarity** The heatmap reveals consistently high cosine similarity scores across all languages, ranging from 0.83 to 0.95. Portuguese exhibits the highest score at 0.95, while German, Japanese, and Turkish have slightly lower scores, around 0.83 to 0.84. This observation suggests that cosine similarity primarily focuses on the overlap of words and phrases between the original and translated texts. The high scores indicate that, despite being translated, the texts retain a

significant portion of their original semantic content. This implies a relatively high level of surface-level similarity across languages. However, relying solely on cosine similarity might be misleading. While it effectively captures lexical overlap, it may not account for subtle differences in meaning or grammatical structure, which are crucial for a comprehensive semantic evaluation. Therefore, while the high scores are encouraging, they should be interpreted with caution.

2. **BLEU** In contrast to cosine similarity, BLEU scores are notably lower, ranging from 0.08 to 0.44. Japanese exhibits the lowest score at 0.08, while Italian has the highest at 0.44. BLEU is more sensitive to word order and grammatical accuracy than cosine similarity. The low scores suggest that the translation process introduces variations in word order and grammar, affecting BLEU's assessment of similarity. This implies that the translated texts, while semantically similar, might deviate significantly in their linguistic structure compared to the originals. BLEU's sensitivity to word order makes it a valuable metric for evaluating the fluency and grammatical correctness of translations. Its low scores in this context highlight the challenges of preserving these aspects during translation.
3. **ROUGE-L and METEOR** ROUGE-L and METEOR scores fall between the extremes of cosine similarity and BLEU, with ROUGE-L ranging from 0.39 to 0.67 and METEOR ranging from 0.42 to 0.72. These metrics consider both lexical overlap and semantic similarity, providing a more balanced assessment. Their intermediate scores suggest that while the translations might not perfectly match the original texts in terms of word order and grammar, they do preserve a considerable amount of semantic meaning. ROUGE-L and METEOR offer a more nuanced view of translation quality, capturing both surface-level and deeper semantic relationships. Their moderate scores indicate a reasonable balance between semantic preservation and linguistic variation across languages.

3.11.7 Translation Services Comparison

We have now a collection of four box plots, each representing the distribution of a specific similarity metric (Cosine Similarity, BLEU, ROUGE-L F1, and METEOR) for translations produced by two different services: Google Translate and DeepL. This visualization

is instrumental in comparing the performance of these translation services based on how well their translations preserve similarity to the original text, as measured by these four metrics.

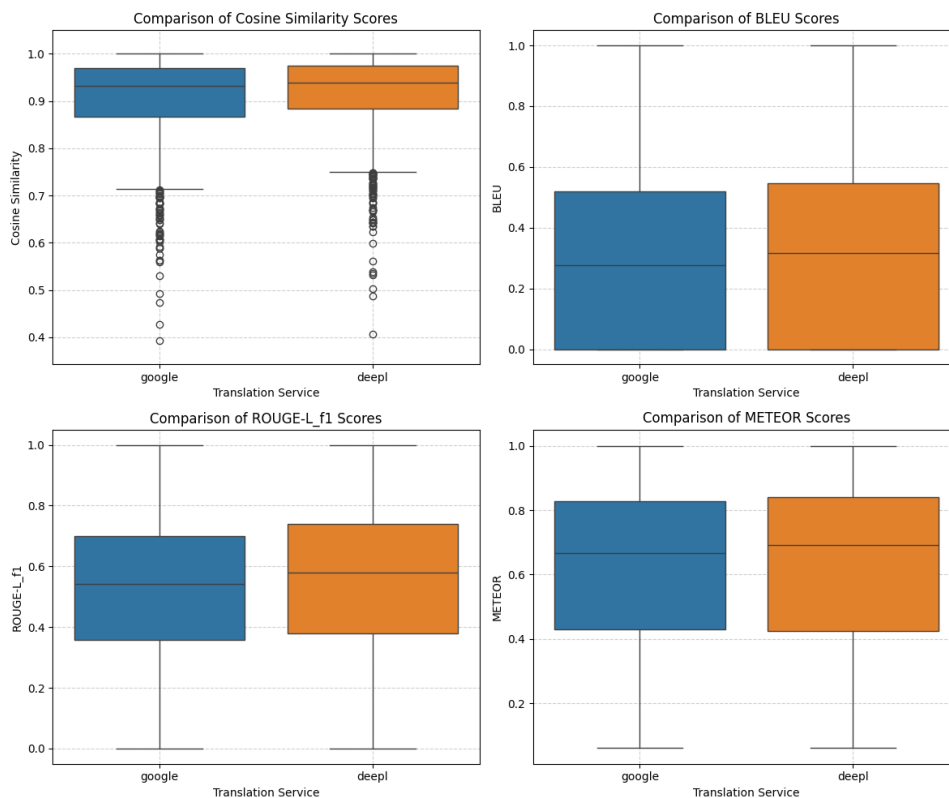


Figure 3.7: Box plots of the data distribution of a specific metric based on the translation service.

Box plots are a powerful tool for visualizing the distribution of data, as they illustrate the median, quartiles, minimum, maximum, and potential outliers. This allows us to see not only the central tendency of the data but also its spread and any anomalies. In this context, each box plot provides insight into how consistently each translation service maintains textual similarity across different metrics.

Starting with Cosine Similarity, we observe that both Google Translate and DeepL generally produce translations with high similarity scores. For Google Translate, the minimum score is around 0.72, with the median at 0.95, and the 75th percentile around 0.98. However, there are outliers below the minimum, indicating instances where Google’s translations result in significantly lower similarity scores. This suggests that while Google Translate typically performs well, there are cases where its performance is inconsistent.

DeepL shows a similar distribution but with a slightly higher minimum score of around 0.75, suggesting that DeepL might be slightly more consistent in maintaining higher similarity scores.

The BLEU scores, on the other hand, are notably lower than those of Cosine Similarity, with Google Translate’s minimum and 25th percentile both at 0, and the median at 0.3. DeepL’s scores are slightly higher, indicating that DeepL’s translations tend to have better word-level overlap with the original text. The lower BLEU scores suggest that this metric is stricter, focusing more on precise word matches and grammatical accuracy. The presence of very low scores for Google Translate indicates that there are instances where the translations have very little overlap with the original text, highlighting the challenges in preserving word order and grammar during translation.

ROUGE-L F1 scores fall between those of Cosine Similarity and BLEU, suggesting a balance between semantic and word-level similarity. Both Google Translate and DeepL have minimum scores at 0, but DeepL’s interquartile range is higher, indicating better performance in preserving both meaning and structure. This suggests that while translations may not perfectly match the original text in terms of word order, they do retain a considerable amount of semantic meaning.

METEOR scores are similar to ROUGE-L F1, showing a balance between meaning and structure. DeepL consistently shows marginally better performance compared to Google Translate, but the differences are less pronounced than with BLEU or ROUGE-L F1. The minimum scores at 0 for both services suggest that, like ROUGE-L F1, METEOR can be sensitive to cases where the translations don’t align well with the original text’s structure.

In conclusion, these box plots provide a comprehensive view of how different similarity metrics align with the performance of Google Translate and DeepL. DeepL consistently performs slightly better than Google Translate across most metrics, particularly in terms of BLEU and ROUGE-L F1. This suggests that DeepL’s translations are more reliable in preserving both the meaning and structure of the original text. Cosine Similarity shows generally high scores for both services, but Google Translate has more outliers, indicating potential inconsistencies. BLEU’s lower scores highlight its stricter nature, focusing on precise word matches and grammatical accuracy. ROUGE-L F1 and METEOR provide a

balanced view of similarity, considering both meaning and structure, with DeepL showing marginally better performance.

Chapter 4

Discussion of Results

4.1 Introduction

The results obtained in this study provide crucial insights into the alignment between computational similarity metrics and human perception. By analyzing various similarity measures, including BLEU, ROUGE- L_{f1} , METEOR, and a cosine similarity-based score, we aimed to assess their effectiveness in evaluating paraphrased and retranslated texts. Additionally, we explored the robustness of these metrics under different transformation processes and their consistency across translation services. This chapter provides an in-depth discussion of our findings, highlighting strengths, limitations, and broader implications.

4.2 Analysis of Correlations with Human Perception

One of the key aspects of our evaluation was determining how well different computational metrics correlate with human ratings. Our findings indicate that METEOR exhibited the highest correlation with human judgments, followed by ROUGE- L_{f1} and BLEU. The cosine similarity score demonstrated the lowest correlation, suggesting that it may not fully capture the nuances of human perception.

These results suggest that METEOR is more sensitive to semantic similarity, likely due to its incorporation of stemming, synonym matching, and paraphrase detection. ROUGE- L_{f1} , while effective, remains more structure-dependent than METEOR. BLEU,

on the other hand, relies heavily on exact n-gram matching, leading to lower recall when significant paraphrasing occurs. The weaker correlation of cosine similarity suggests that embedding-based approaches, while useful in some contexts, may not fully align with human similarity perception when dealing with short text segments.

4.3 Evaluation of Metric Performance

To further understand the reliability of these metrics, we analyzed their classification performance using precision, recall, and F1-score. METEOR again stood out, achieving the highest F1-score, indicating a balanced trade-off between precision and recall. BLEU, despite its high precision, suffered from very low recall, highlighting its inability to detect paraphrased content effectively. This suggests that BLEU is overly conservative in its similarity assessment, penalizing even minor rewording, making it less suitable for assessing text modifications that retain semantic meaning.

ROUGE- L_{f1} provided a more balanced performance, demonstrating its usefulness in capturing similarities beyond exact word matches. The cosine similarity metric showed inconsistent results, indicating that its effectiveness depends largely on the nature of the text transformations. Its fluctuating precision suggests that it is susceptible to misclassifications, possibly due to its sensitivity to minor lexical variations rather than underlying semantic meaning.

4.4 Impact of Transformation Processes

The robustness of similarity metrics was tested against different transformation techniques, including grammar modifications, contextual paraphrasing, and BERT-based lexical replacements. The results indicate that:

- METEOR and ROUGE- L_{f1} remained relatively stable across transformations, reinforcing their ability to capture paraphrased and restructured text.
- BLEU showed a significant drop in similarity scores after paraphrasing and BERT-based replacements, confirming its dependency on lexical overlap.
- Cosine similarity exhibited erratic behaviour, often overestimating similarity when

minor changes were made while underestimating similarity for structurally different but semantically equivalent texts.

The implications of these findings suggest that similarity metrics designed for machine translation evaluation may not be directly applicable to paraphrase detection without modifications. Metrics like BLEU, which prioritize exact matches, fail to capture meaning-preserving transformations, whereas METEOR’s reliance on synonym handling provides a more robust measure of similarity.

4.5 Impact of Linguistic Distance on Similarity Scores

One of the most striking patterns emerging from the analysis is the systematic variation in similarity scores across languages. Languages closely related to English, such as Italian, Spanish, and Portuguese, consistently achieved higher similarity scores, whereas languages more structurally distant from English, such as Japanese and Turkish, recorded notably lower scores. This trend was observed across all metrics, but it was particularly evident in BLEU and ROUGE-L, which rely heavily on lexical and syntactic overlap.

The reason for this discrepancy lies in fundamental linguistic differences. Word order and sentence structure play a significant role; English follows a subject-verb-object (SVO) structure, whereas Japanese and Turkish use a subject-object-verb (SOV) order. During translation, sentences undergo major syntactic rearrangements, which significantly lower BLEU and ROUGE scores. Additionally, morphological complexity contributes to these differences. Japanese and Turkish are agglutinative languages, meaning that grammatical relationships are encoded through affixes and extensive inflection, rather than separate words as in English. This results in mismatches at the token level, reducing n-gram overlap and lowering metric scores. Word segmentation also plays a role, as Japanese does not use spaces to separate words, leading to more fragmented translations when tokenized for similarity scoring, further reducing BLEU and ROUGE-L precision.

These linguistic differences explain why BLEU scores for Japanese were as low as 0.08, while for Italian they reached 0.44. In contrast, Cosine Similarity exhibited smaller variations across languages because it evaluates meaning based on vector embeddings rather than exact lexical overlap. However, even in cosine similarity, Japanese and Turkish

showed slightly lower scores due to semantic shifts in translation.

4.6 Comparison of Translation Services: Google vs. DeepL

An important aspect of this study was assessing the consistency of similarity metrics across different machine translation services. By comparing Google Translate and DeepL, we observed:

- DeepL translations tended to preserve sentence structure more faithfully, leading to higher similarity scores across all metrics.
- Google Translate introduced greater variability in phrasing, causing more fluctuations in similarity scores.
- The impact of retranslation was more pronounced in Google Translate, where back-translated sentences sometimes diverged significantly from the original.

These findings suggest that DeepL’s approach to translation is more conservative, maintaining greater lexical consistency, while Google Translate’s transformations introduce more paraphrasing. This discrepancy impacts similarity metrics, particularly those dependent on surface-level text matching. Future studies could further explore the effects of different translation models on text similarity assessments.

4.7 Practical Implications and Future Directions

Our findings highlight the limitations of traditional similarity metrics in capturing human-perceived semantic equivalence. METEOR emerges as the most reliable measure, making it a strong candidate for applications requiring paraphrase detection or plagiarism assessment. However, even METEOR has its weaknesses, particularly in handling translations that involve syntactic restructuring rather than direct synonym replacements.

Future work could involve:

- Incorporating contextual embeddings (e.g., BERT, SBERT) for improved semantic similarity detection.

- Exploring hybrid approaches that combine multiple metrics to leverage their strengths.
- Evaluating the impact of translation quality on similarity assessments across more languages and translation models.

The findings confirm that traditional similarity metrics vary in their effectiveness, with METEOR emerging as the most robust for evaluating paraphrased and retranslated text. The evaluation also underscores the limitations of BLEU and cosine similarity for assessing meaning-preserving transformations. The comparison between Google Translate and DeepL further highlights how translation-induced variations can impact similarity assessments. These insights contribute to a deeper understanding of how well computational metrics align with human perception and inform future work in text similarity evaluation.

Chapter 5

Conclusion

This thesis set out to evaluate the reliability of computational similarity metrics in assessing textual similarity compared to human judgment. By applying layered paraphrasing, translation, and retranslation processes, we analyzed how well different metrics—Cosine Similarity, BLEU, ROUGE-L, and METEOR—aligned with human-perceived similarity.

The results indicate that METEOR consistently outperformed other metrics in terms of correlation with human judgments. It demonstrated the highest correlation coefficients and provided a balanced assessment of similarity by integrating lexical, semantic, and ordering-based considerations. ROUGE-L also exhibited strong performance, capturing some structural similarities that BLEU and Cosine Similarity overlooked. BLEU, while widely used in machine translation evaluation, struggled significantly with paraphrased texts due to its reliance on exact n-gram matches. Cosine Similarity, though useful in embedding-based representations, showed weaker alignment with human perception.

The human evaluation phase highlighted key challenges in textual similarity assessment. Annotators often disagreed on sentence similarity, especially for sentences modified through translation and retranslation. To handle annotator disagreement, we computed the mean rating for each sentence pair, ensuring that extreme opinions did not overly influence the results. This aggregation method provided a more stable representation of human perception, aligning better with the goal of assessing semantic equivalence.

A significant observation from the multilingual analysis was that translation across languages from different linguistic families, such as Japanese and Turkish, led to system-

atically lower similarity scores. This suggests that certain languages introduce greater structural and lexical variations, making it more difficult for traditional similarity metrics to detect equivalence accurately. This finding highlights the importance of developing cross-linguistic similarity measures that account for typological differences.

The comparison of translation services revealed that DeepL generally preserved textual similarity better than Google Translate, particularly for Romance languages. However, variations between translation services were not as pronounced as the impact of language choice, suggesting that the choice of target language plays a more substantial role in similarity score fluctuations than the specific translation service used.

Overall, this study underscores the limitations of current similarity metrics when applied to paraphrased, translated, and retranslated texts. While METEOR and ROUGE-L demonstrated the strongest alignment with human perception, their performance varied across linguistic contexts. Future work should focus on developing improved similarity measures that integrate syntactic, semantic, and contextual features more effectively, potentially leveraging recent advancements in transformer-based models and multilingual embeddings.

In conclusion, computational similarity metrics provide a useful but imperfect approximation of human similarity judgments. Their reliability depends on the type of textual transformation applied, the linguistic characteristics of the text, and the evaluation criteria used. A multi-metric approach that combines different measures may offer the best solution for detecting semantic similarity in diverse textual contexts.

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